



Chinook Pedicab Production Manual

1. Overview

The Chinook is a high-performance long-wheelbase (LWB) pedicab designed for urban passenger transport. Its primary innovation is a structural bamboo-composite frame that combines sustainable materials with high-tensile strength. The design supports a 250kg-300kg payload and is optimized for both artisan hand-fabrication and scalable industrial production.

Key Dimensions:

- Total Vehicle Length: 3050 mm (120")
- Wheelbase (Axle-to-Axle): 1850 mm (72.8")
- Main Chassis Rail: 2700 mm (106.3")
- Passenger Legroom: 800 mm (31.5")

Figure 1: Chassis Rail Construction Detail showing bamboo dowel reinforcement and core integration.

2. Materials & Sourcing

The Chinook utilizes a hybrid of natural bamboo, carbon fiber, and epoxy resins.

Local Sourcing (Vancouver):

- **Coolite Bamboo (917 East Hastings St):** Supplier for heat-treated bamboo poles (to be split into strips) and artisan rattan weaving for seating.
- **Industrial Plastics & Paints (Richmond/PoCo):** Supplier for laminating epoxy resins, unidirectional carbon fiber fabric, and industrial heat-shrink tubing.

Figure 2: Passenger Comfort details showing the Artisan Rattan Weave and BB-03 bench frame construction.

3. Handmade Method (Artisan Lamination)

The handmade method focuses on accessibility and cost-efficiency for small-batch production (1-2 units). It maximizes bamboo's natural tensile strength through manual lamination.

3.1 Lamination Jigs ("Blue Boards") Jigs are used as moulds to shape the curved components. Cut templates from 18mm plywood or MDF (cut two of each for parallel lamination):

- **BB-01:** Main Chassis Rail (2400 mm) - Primary structural curve.
- **BB-02:** Wheel Arch Support (1200 mm) - Rear axle mounting curve.
- **BB-03:** Passenger Bench Frame (1300 mm) - Seat structure support.
- **BB-04:** Steering Column Strut (900 mm) - Front fork connection.

Figure 3: Fabrication setups for Wheel Arch (BB-02) and Passenger Bench Frame (BB-03) using 18mm plywood templates.

3.2 The 12-Layer Schedule To achieve the 60mm structural thickness:

1. **Preparation:** Split bamboo poles into 5mm strips.
2. **Lamination:** Glue 12 layers of strips into the jigs.
3. **Reinforcement:** Integrate unidirectional carbon fiber cores between layers for added stiffness
4. **Compression:** Wrap assembly in industrial heat-shrink tubing to provide even, 360-degree pressure during the 24-36 hour cure cycle.

4. Mass Production Method (Industrial Scaling)

For higher volumes, the manual transitions to pre-fabricated composite components and standardized assembly.

- **Standardized Tubes:** Replace hand-laminated rails with 60mm carbon-wrapped bamboo composite tubes.
- **CNC Machining:** Use CNC-cut jig systems for faster assembly and higher precision.
- **Modular Components:** Transition to standardized sub-assemblies for the steering column and rear axle interface.

5. Engineering Specifications

The Chinook is engineered for heavy-duty service and safety.

- **Chassis:** Bamboo-Carbon Composite (65mm thickness at critical nodes).
- **Payload Capacity:** Certified for 300kg+ service.
- **Motor Assist:** Designed for 500W mid-drive or hub motor integration.
- **Drive Interface:** 5-bolt hub interface for high-torque demands.
- **Braking:** Hydraulic disc brakes on all three wheels.
- **Validation:** FEA (Finite Element Analysis) verified for bamboo-to-metal interfaces at the carbon-reinforced axle points.

Figure 4: Steering Column Fabrication showing the BB-04 template and composite curves.